SPRiF: Spectral Memory and Scalar Reset for Spiking Neurons

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***Abstract*—Temporal processing in spiking neural networks requires neurons to retain input history while repeatedly generating spikes. When a membrane carries both functions, reset enters the same recurrence as the stored history. We present SPRiF, a five-state neuron that couples a constrained spectral memory and an auxiliary linear filter to a scalar resettable membrane. The spectral block contains one real decay mode and a damped rotational pair. Two learned readouts provide a direct membrane drive and a filtered drive with a one-step delay. A spike subtracts one fixed threshold from the membrane and leaves the four memory and filter coordinates unchanged by direct local reset. The parameterization uses eleven trainable dynamics coefficients per neuron. We derive the spectral and effective drive kernels, establish controllability of the three-dimensional spectral block, and prove boundedness of the local states for bounded current. We also specify the reset identity and its boundary under recurrent feedback. The empirical protocol covers five temporal classification benchmarks, two controls with matched state and coefficient counts, and a recurrent spike-intervention task. Measurements for this parameterization are pending; the available historical results are identified separately.**

***Index Terms*—Spiking neural networks, temporal processing, spectral memory, neuron dynamics, scalar reset.**

I. Introduction

SPIKING neural networks (SNNs) process temporal signals through stateful neurons and discrete output events. In sequence classification, a neuron must combine recent evidence with context accumulated much earlier. Its temporal representation must remain useful while the network continues to emit spikes. The way state variables carry memory and respond to each spike is therefore part of the computational design.

A leaky integrate-and-fire (LIF) neuron uses a scalar membrane to accumulate current, determine firing, and receive reset. Unrolling its recurrence exposes both input history and reset history in the same decaying trace. This arrangement is compact, but it ties the retained representation to the neuron's discharge events. Adaptive spiking neurons extend this state with adaptation variables [1], [2]. Resonant neurons introduce oscillatory temporal responses [3]–[5], and compartmental models distribute integration across coupled coordinates [6], [7]. These approaches motivate examining which states carry temporal information and how that information reaches the spike-generating membrane.

Separating memory from output reset is already supported by several neuron and network constructions. PMSN organizes linear hidden compartments and an output compartment with firing and reset, and represents hidden dynamics through spectral coordinates [8]. DMP-SNN maintains a compact memory shared within each layer [9]. The design question addressed here is more specific: how should a fixed, small state budget organize monotone retention, damped phase responses, and their coupling to a resettable membrane? State count alone does not answer this question because models with the same number of coordinates can have different driven modes and reset paths.

SPRiF assigns three coordinates to a spectral block, one to an auxiliary filter, and one to the membrane. The spectral block contains a real exponential mode and a real-valued representation of a complex-conjugate pair. One learned projection drives the membrane directly. A second projection enters the auxiliary recurrence, whose previous value also drives the membrane. The resulting four-dimensional memory and filter pathway is linear for a specified current sequence. The membrane emits a binary spike and receives a scalar subtractive reset.

This organization supplies a tractable connection between the state equations and their temporal kernels. The decay and rotation constraints place the spectral poles inside the unit disk, while the auxiliary filter adds a learned pole and a delayed path. All three spectral coordinates are controllable under the stated open parameter ranges. Direct reset changes the scalar membrane alone. In recurrent layers, the resulting spike still affects future currents, so the local reset identity and network responses must be examined at different levels.

The analysis connects SPRiF's constrained modes and auxiliary delay to effective kernels, controllability, and conditional state bounds. The evaluation tests these choices through two controls with the same state and dynamics-coefficient counts, controlled baselines, initialization sensitivity, and resource measurements. It covers S-MNIST, PS-MNIST, QTDB, GSC, SHD, and Spike-Intervention Delayed Match-to-Sample (SIDMS). Section V separates historical measurements from the pending results for the current model.

II. Related Work

A. Adaptive, Resonant, and Compartmental Neurons

Adaptation augments membrane integration with another dynamical variable or threshold process. ASRNN and adLIF use such dynamics to improve temporal processing [1], [2]. Their temporal responses depend on how adaptation interacts with membrane voltage and output spikes. SPRiF instead specifies a linear memory and filter pathway upstream of a scalar spiking membrane.

Resonate-and-fire neurons represent frequency-sensitive responses through coupled states [3]. BRF combines resonant dynamics with mechanisms for stable spiking computation [4]. D-RF uses dendritic resonant recurrences, a weighted somatic readout, and a spike-history-dependent threshold [5]. Its dendritic recurrence also excludes a subtractive output-spike reset. Relative to that organization, SPRiF uses one real mode and one rotational pair, filters a second spectral readout through an

auxiliary state, and drives a membrane with its own leak and scalar reset. The shared resonant and reset-separated elements are prior mechanisms; the comparison concerns their particular state and output organization.

TC-LIF couples dendritic and somatic states for sequential processing [6]. Dendritic heterogeneity distributes time constants across compartments [7]. These studies establish the usefulness of multiple temporal coordinates. The controls in Section IV-C therefore retain SPRiF's state and coefficient counts while changing either its spectral modes or a specific memory reset path.

B. Linear Memory Pathways and Spectral Realizations

PMSN separates $n-1$ linear hidden compartments from a spike-generating output compartment [8]. Its hidden transition is diagonalized and discretized to obtain a spectral recurrence and a convolutional readout. SPRiF's four memory coordinates also admit a linear hidden-state representation. Its prescribed real block structure and auxiliary filter select a particular family of poles and couplings within that broader class of constructions.

The final PMSN formulation introduces temporal-decoupled reset (TDR). Its output accumulator uses unit retention, and the discharge is an integer multiple of the threshold determined by a floor operation. The equivalence of its serial and parallel TDR forms uses a nonnegative output current. SPRiF retains a learned membrane leak, allows signed readout drive, and subtracts one threshold per emitted binary spike. Table I states these differences at the level of equations and assumptions. They do not by themselves establish an accuracy or efficiency advantage.

DMP-SNN places a compact shared memory at the layer level [9]; SPRiF maintains four memory and filter values within each neuron. SpikingSSMs also combine structured state-space processing with sparse spiking outputs [10]. These connections make memory organization, state storage, and the full processing path relevant comparison axes.

C. Reset Formulations and Training

NiSNN-A computes an input accumulation in matrix form and estimates the spike contribution to reset before producing the final NiLIF output [11]. Its noniterative formulation uses an approximation to the unknown spike history. SPRiF uses a sequential threshold and reset recurrence; the kernels derived below characterize its linear drive, not a noniterative replacement for its spikes.

IML-Spikeformer combines input-aware multilevel firing with attention using hierarchical decay masks for speech processing [12]. STDL changes the training dependency structure by spatial and temporal decoupling under a memory budget [13]. These works address output encoding, architecture, and training rather than the same local neuron substitution. SPRiF retains binary firing and training by surrogate-gradient backpropagation through time (BPTT). Its intrinsic state storage and the activation memory required by BPTT are reported separately.

III. SPRiF Neuron: Model and Analysis

A. From a Reset-Coupled Trace to Separate State Roles

To fix the timing convention, let v_t be a LIF membrane immediately after the current reset. Its candidate, spike, and stored voltage are

$$\widetilde{v}_t = \beta v_{t-1} + I_t, \qquad z_t = H(\widetilde{v}_t - \theta), \qquad v_t = \widetilde{v}_t - \theta z_t, \tag{1}$$

where $0 < \beta < 1$, $\theta > 0$, and $H(a) = 1$ for $a \geq 0$ and zero otherwise. Unrolling gives

$$v_t = \beta^{t+1} v_{-1} + \sum_{j=0}^{t} \beta^{t-j} I_j - \theta \sum_{j=0}^{t} \beta^{t-j} z_j. \tag{2}$$

Both histories enter the same state. SPRiF places the linear representation of current history upstream of the membrane so that direct local reset does not alter its coordinates.

The three spectral states describe the input trace. The auxiliary state integrates a second learned spectral projection and adds a delayed path. The membrane combines these drives, retains its own previous post-reset value, and produces the spike. Each role follows from its recurrence; the construction does not require all memory time constants to be longer than the membrane time constant.

B. Complete Forward Definition

For an input vector $\mathbf{s}_t$ and a recurrent spiking layer, the current vector is

$$\mathbf{I}_t = W_{\text{in}} \mathbf{s}_t + W_{\text{rec}} \mathbf{z}_{t-1} + \mathbf{b}. \tag{3}$$

The recurrent term is omitted for a feedforward layer. The equations below concern one neuron and suppress its index. Its current I_t is the corresponding component of $\mathbf{I}_t$, and it stores $\mathbf{x}_t \in \mathbb{R}^3$, $h_t \in \mathbb{R}$, and $u_t^0 \in \mathbb{R}$:

$$\begin{cases} \mathbf{x}_t = A\mathbf{x}_{t-1} + B I_t, \\ h_t = \eta_1 h_{t-1} + \bar{\mathbf{g}}_1^\top \mathbf{x}_t, \\ \widetilde{u}_t^0 = \eta_0 u_{t-1}^0 + h_{t-1} + \mathbf{g}_0^\top \mathbf{x}_t, \\ z_t = H(\widetilde{u}_t^0 - \theta), \\ u_t^0 = \widetilde{u}_t^0 - \theta z_t. \end{cases} \tag{4}$$

The transition and input vector are

$$A = \begin{bmatrix} \alpha & 0 & 0 \\ 0 & \rho\cos\omega & -\rho\sin\omega \\ 0 & \rho\sin\omega & \rho\cos\omega \end{bmatrix}, \qquad B = \begin{bmatrix} 1-\alpha \\ 1-\rho \\ 0 \end{bmatrix}. \tag{5}$$

The two directly learned readouts form

$$\bar{G} = \begin{bmatrix} \mathbf{g}_0^\top \\ \bar{\mathbf{g}}_1^\top \end{bmatrix} \in \mathbb{R}^{2\times 3}. \tag{6}$$

For finite raw parameters, the constraints are enforced by

$$\begin{aligned} &\alpha = \sigma(\alpha_{\text{raw}}), \qquad \rho = \sigma(\rho_{\text{raw}}), \\ &\omega = \pi\sigma(\omega_{\text{raw}}), \quad \eta_i = \sigma(\eta_{i,\text{raw}}), \quad i \in \{0,1\}. \end{aligned} \tag{7}$$

Here σ is the logistic sigmoid. Thus $\alpha, \rho, \eta_i \in (0,1)$ and $\omega \in (0,\pi)$. The six entries of $\bar{G}$ are unconstrained real coefficients. The threshold is a fixed task setting.

TABLE I
EQUATION-LEVEL COMPARISON WITH PMSN. PMSN ENTRIES REFER TO ITS HIDDEN SPECTRAL FORM AND FINAL TDR FORMULATION [8].

Aspect	PMSN	SPRiF
State organization	Adjustable linear hidden compartments and one output compartment.	Three spectral coordinates, one auxiliary filter, and one membrane.
Memory kernel	Hidden transition is diagonalized and discretized; input and output projections define a spectral convolution.	A prescribed real decay and rotational block; direct and delayed auxiliary readouts define Eq. (15).
Output dynamics	Final TDR uses unit retention and a threshold-quantized discharge.	Learned leak η_0 and a single-threshold subtraction for each binary spike.
Parallel-reset assumption	Serial/parallel TDR equivalence uses nonnegative current after rectification.	The sequential update accepts signed drive; no parallel-reset equivalence is asserted.
Direct local reset	Output compartment receives reset; hidden compartments remain linear.	Only u_t^0 receives reset; $(\mathbf{x}_t, h_t)$ remain unchanged at that event.

The update order is essential. Compute $\mathbf{x}_t$ first, then compute h_t and $\widetilde{u}_t^0$ using that new spectral state and the stored h_{t-1}. Emit z_t and store the scalar post-reset membrane. Replacing h_{t-1} by h_t would remove the specified delay and change the model. Unless an explicit state conversion is used, initial states are zero and states are reset between independent sequences. State handling at truncated sequence boundaries is specified in Section IV-B.

C. Absorbed Auxiliary Scaling

The coefficient of h_{t-1} in Eq. (4) is fixed at one because its amplitude is already represented by the second readout row. To show this, consider the former scalar-reset parameterization with auxiliary coordinate u_t^1:

$$\begin{aligned} u_t^1 &= \eta_1 u_{t-1}^1 + \mathbf{g}_1^\top \mathbf{x}_t, \\ \widetilde{u}_t^0 &= \eta_0 u_{t-1}^0 + \kappa u_{t-1}^1 + \mathbf{g}_0^\top \mathbf{x}_t. \end{aligned} \tag{8}$$

In this conversion only, κ denotes a time-independent legacy coupling. Define

$$h_t = \kappa u_t^1, \qquad \bar{\mathbf{g}}_1 = \kappa \mathbf{g}_1, \qquad h_{-1} = \kappa u_{-1}^1. \tag{9}$$

Multiplication of the first recurrence by κ and substitution into the second gives Eq. (4). No division is required: at $\kappa = 0$, $h_{-1} = 0$ and $\bar{\mathbf{g}}_1 = \mathbf{0}$ preserve the membrane and spike sequence. With corresponding initial states, the equality extends through a recurrent network because identical previous spikes give identical next currents.

This conversion assumes a free readout row and no separately required auxiliary output or amplitude constraint. SPRiF directly learns $\bar{\mathbf{g}}_1$; κ is not a parameter of its current forward model. The auxiliary state remains present. Absorbing a coupling preserves fixed-parameter forward outputs for the scalar-reset formulation, while directly optimizing the product changes the optimization coordinates. Separate legacy penalties or constraints require an explicit new treatment. Historical training results therefore retain the model identities stated in Section V-A.

D. Spectral Basis, Controllability, and Effective Kernels

The poles of the spectral transition are

$$\operatorname{spec}(A) = \{\alpha, \rho e^{\mathrm{i}\omega}, \rho e^{-\mathrm{i}\omega}\}. \tag{10}$$

For a unit impulse $I_0 = 1$, zero state at $t = -1$, and zero later current,

$$A^k B = \begin{bmatrix} (1-\alpha)\alpha^k \\ (1-\rho)\rho^k \cos(k\omega) \\ (1-\rho)\rho^k \sin(k\omega) \end{bmatrix}, \qquad k \geq 0. \tag{11}$$

The first component gives monotone exponential retention. The other two provide phase-shifted responses with a common damping factor and frequency. Across neurons, learned decays, frequencies, and projections can distribute these responses over different temporal scales.

Proposition 1 (Driven spectral coordinates): For $\alpha, \rho \in (0, 1)$ and $\omega \in (0, \pi)$, the pair (A, B) in Eq. (5) is controllable.

Proof: Let $\mathcal{C} = [B, AB, A^2B]$. Expanding its determinant gives

$$\begin{aligned} \det \mathcal{C} = (1-\alpha)(1-\rho)^2 \rho \sin\omega \\ \times (\alpha^2 - 2\alpha\rho\cos\omega + \rho^2). \end{aligned} \tag{12}$$

Every factor is positive, since the last factor equals $|\alpha - \rho e^{\mathrm{i}\omega}|^2$. Thus $\mathcal{C}$ has rank three. ∎

The proposition concerns the spectral block. Nearly zero $\sin\omega$ or small input factors can still produce poor numerical conditioning. It also does not imply that every mode is visible in a learned output: a readout can suppress a mode. Numerical singular values and readout scales are therefore relevant diagnostics.

Under the same impulse, the auxiliary response is

$$h_k = \sum_{\ell=0}^{k} \eta_1^{k-\ell} \bar{\mathbf{g}}_1^\top A^\ell B. \tag{13}$$

Write the membrane drive as

$$d_t = \mathbf{g}_0^\top \mathbf{x}_t + h_{t-1}. \tag{14}$$

Its impulse kernel is

$$\begin{aligned} k_d[k] = \mathbf{g}_0^\top A^k B \\ + \sum_{\ell=0}^{k-1} \eta_1^{k-1-\ell} \bar{\mathbf{g}}_1^\top A^\ell B, \end{aligned} \tag{15}$$

where the sum is zero at $k = 0$. The direct path responds immediately, whereas the auxiliary path first contributes at $k =$

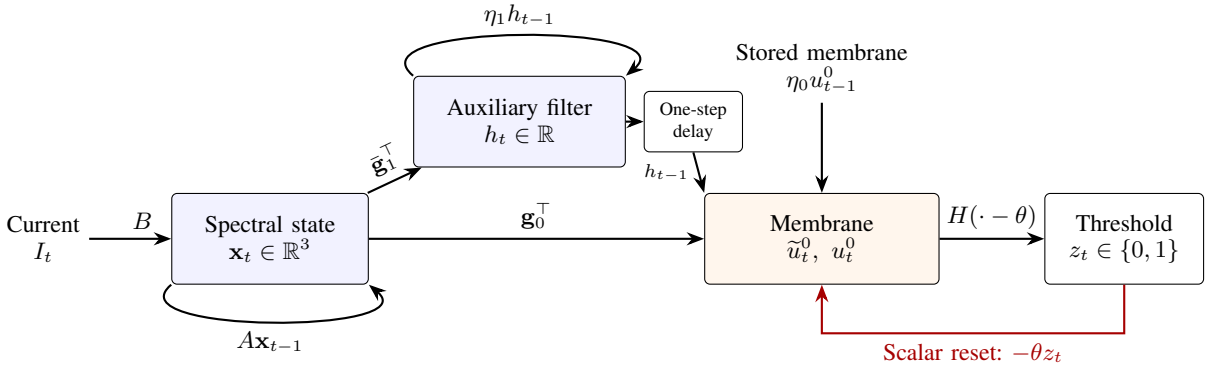


Fig. 1. SPRiF state organization. The spectral and auxiliary recurrences carry four memory and filter values. The membrane candidate combines the current spectral readout, the previous auxiliary value, and the previous post-reset membrane. The current reset acts on the scalar membrane. This diagram specifies equations rather than measured trajectories.

1. If ξ denotes a formal unit-delay variable, the same transfer function is

$$\mathcal{K}_d(\xi) = \left(\mathbf{g}_0^\top + \frac{\xi}{1-\eta_1\xi}\bar{\mathbf{g}}_1^\top\right)(I_3 - \xi A)^{-1}B. \tag{16}$$

For a membrane with its spike reset suppressed, an additional factor $(1-\eta_0\xi)^{-1}$ gives its linear response. The actual stored membrane instead satisfies the exact expansion

$$\begin{aligned} u_t^0 = \eta_0^{t+1}u_{-1}^0 + \sum_{j=0}^{t}\eta_0^{t-j}d_j \\ - \theta\sum_{j=0}^{t}\eta_0^{t-j}z_j. \end{aligned} \tag{17}$$

Thus Eq. (16) describes the linear drive, while the spike history remains in the membrane recurrence.

When an oscillatory pole has nonzero residue in the output kernel, a finite sum of modes with real positive poles cannot reproduce that kernel for all lags: equal rational transfer functions must have the same uncancelled poles. This specifies a distinction between the selected kernel families, not a universal advantage on a finite task. The real-mode control in Section IV-C tests its empirical relevance with the same local capacity.

E. Conditional State Bounds

For fixed parameters and a specified current sequence, define $\mathbf{m}_t = [\mathbf{x}_t^\top, h_t]^\top$. The memory pathway is

$$\mathbf{m}_t = F\mathbf{m}_{t-1} + LI_t, \tag{18}$$

with

$$F = \begin{bmatrix} A & \mathbf{0} \\ \bar{\mathbf{g}}_1^\top A & \eta_1 \end{bmatrix}, \qquad L = \begin{bmatrix} B \\ \bar{\mathbf{g}}_1^\top B \end{bmatrix}. \tag{19}$$

The block-triangular structure gives

$$\operatorname{spec}(F) = \{\alpha, \rho e^{\mathrm{i}\omega}, \rho e^{-\mathrm{i}\omega}, \eta_1\}. \tag{20}$$

All poles lie strictly inside the unit disk. This spectral property does not require $\|F\|_2 < 1$: its learned readout can produce transient amplification.

Proposition 2 (Bounded local states): Assume fixed finite readouts, finite initial states, $\theta > 0$, the ranges in Eq. (7), and $|I_t| \le M$ for all $t \ge 0$. Then $\mathbf{x}_t$, h_t, and u_t^0 in Eq. (4) remain bounded.

Proof: Let $q = \max(\alpha, \rho) < 1$. Since the rotational block is ρR_ω, $\|A^k\|_2 = q^k$. Unrolling the spectral recurrence gives

$$\|\mathbf{x}_t\|_2 \le q^{t+1}\|\mathbf{x}_{-1}\|_2 + \|B\|_2 M\frac{1-q^{t+1}}{1-q}. \tag{21}$$

One uniform bound is $X = \|\mathbf{x}_{-1}\|_2 + \|B\|_2 M/(1-q)$. The auxiliary recurrence then yields

$$|h_t| \le \eta_1^{t+1}|h_{-1}| + \|\bar{\mathbf{g}}_1\|_2 X\frac{1-\eta_1^{t+1}}{1-\eta_1}. \tag{22}$$

Put $H_* = |h_{-1}| + \|\bar{\mathbf{g}}_1\|_2 X/(1-\eta_1)$ and $D_* = \|\mathbf{g}_0\|_2 X + H_*$. Then $|d_t| \le D_*$. Using $z_t \in \{0,1\}$ in Eq. (17),

$$|u_t^0| \le \eta_0^{t+1}|u_{-1}^0| + (D_* + \theta)\frac{1-\eta_0^{t+1}}{1-\eta_0}, \tag{23}$$

which is finite. ∎

The bounds describe the local forward states conditional on bounded current. They neither establish a contraction of the nonlinear recurrent network nor guarantee well-conditioned BPTT gradients. Large readouts and decays near one can yield large finite bounds. Coincident poles preserve boundedness but can change transients. Initialization, learned state scales, and recurrent responses are therefore evaluated alongside accuracy.

F. Reset Identity and Recurrent Feedback

Use superscripts $-$ and $+$ for values immediately before and after the current local reset. Equation (4) gives

$$\mathbf{x}_t^+ = \mathbf{x}_t^-, \qquad h_t^+ = h_t^-, \qquad u_t^{0,+} = \widetilde{u}_t^0 - \theta z_t. \tag{24}$$

The reset increments of all four memory and filter coordinates are exactly zero for either spike outcome. This is an algebraic property of the current update, rather than an empirical robustness claim.

At fixed I_t, the memory-to-memory Jacobian is F and contains no local threshold derivative. Its k-step intrinsic

propagation is F^k. In a recurrent layer, however, changing a spike changes a later current by

$$\delta \mathbf{I}_{t+1} = W_{\mathrm{rec}}\, \delta \mathbf{z}_t \tag{25}$$

when the external layer input is held fixed at that step. Later memory states can consequently differ. SI-DMS measures the complete response to membrane overwrite, forced firing, native reset, and recurrent propagation. The local identity specifies which immediate reset path is excluded; it does not isolate every cause of a later accuracy difference.

Single-threshold subtraction also leaves any excess voltage above one threshold in the stored membrane. Binary output is limited to one event per timestep, so a very large positive candidate need not become subthreshold after reset. This convention is part of Eq. (4) and differs from a threshold-quantized or multilevel discharge.

G. Learning and Computational Accounting

The hard threshold is used in the forward pass. For BPTT, its derivative is replaced by the multi-Gaussian surrogate used in the original specification [1]:

$$\begin{aligned}\psi(v) = \gamma\big[&a_0 \mathcal{N}(v-\theta; 0, \sigma_0^2) \\ &- a_1 \mathcal{N}(v-\theta; \mu_1, \sigma_1^2) \\ &- a_1 \mathcal{N}(v-\theta; -\mu_1, \sigma_1^2)\big],\end{aligned} \tag{26}$$

where $\mathcal{N}(\cdot; \mu, \sigma^2)$ is the normalized Gaussian density, $\sigma_0 = 0.5$, $\mu_1 = 0.5$, $\sigma_1 = 3.0$, $a_0 = 1.15$, $a_1 = 0.15$, and $\gamma = 0.5$. The exact gradient treatment of reset and truncation is a protocol field in Section IV-B.

Each neuron stores (x^0, x^1, x^2, h, u^0), for five state values. It learns the five raw variables associated with $\alpha, \rho, \omega, \eta_0, \eta_1$ and six entries of $\bar{G}$, for eleven dynamics coefficients. The count is not a claim of a globally minimal realization. Thresholds are fixed, and synaptic weights, biases, and output parameters are separate.

For a recurrent layer with input width m and hidden width n, a dense input matrix, dense recurrent matrix, and one current bias per neuron contribute

$$N_{\mathrm{layer}} = mn + n^2 + n + 11n. \tag{27}$$

Output and additional architectural parameters are added according to their actual configuration. The persistent local state contains $5n$ scalar values, or $5nb_{\mathrm{val}}$ bytes when each value uses b_{val} bytes. This excludes weights, optimizer buffers, and BPTT activations. Local updates cost $O(n)$ per step; dense synaptic products cost $O(mn+n^2)$. Timing and peak memory require measurements under specified execution conditions.

IV. Experimental Setup

A. Tasks, Encodings, and Data Partitions

The five existing benchmarks span serialized images, physiological signals, dense speech features, and event-based speech. Their roles differ: S-MNIST tests long sequential integration; PS-MNIST disrupts local pixel order; QTDB uses a label at each timestep; GSC classifies a complete feature sequence; and SHD uses sparse cochlear events. Dataset sources define the original data, while the encodings below follow the supplied experimental specification.

S-MNIST and PS-MNIST use MNIST [14] images as 784 scalar pixel inputs. PS-MNIST uses a permutation generated from the run seed and held fixed within that run. Every compared model at the same seed uses the same permutation. Variation across seeds therefore includes permutation variation as well as the training realization. The pixel normalization and the training/validation partition for the new runs remain *TBD*.

QTDB uses the QT Database [15] with the event encoding adopted in ASRNN [1]. Two normalized ECG channels are smoothed and converted into positive and negative first-difference events, yielding four input streams. Six waveform classes are predicted at each labeled timestep. The split must identify the original records or subjects from which windows are derived, so that overlapping or related windows are assigned consistently. Record identifiers, window lengths and overlaps, normalization sources, event thresholds, and the timestep interpretation are *TBD*. These fields are necessary to interpret a new QTDB accuracy.

GSC uses one-second utterances from Speech Commands [16]. The supplied configuration uses 40 mel bins with first- and second-order deltas, producing 120 input features across 101 frames and a 12-class output. Hidden spikes are pooled over time before classification. The dataset release, exact class mapping including any silence or unknown classes, partition counts, audio normalization, and feature extraction constants are *TBD*.

SHD uses the 700-channel cochlear spike trains of Spiking Heidelberg Digits [17], with 20 classes. The supplied configuration accumulates output evidence after a warm-up period. Bin duration, temporal truncation or padding, the warm-up length, output recurrence, validation partition, and handling of variable sequence length are *TBD*.

The new runs use a fixed training/validation/test partition. Where no validation partition is specified, it will be selected from training data before formal training. Validation chooses the checkpoint using a predetermined metric, with the earliest checkpoint resolving a tie; the test set is used for final evaluation. The actual split identifiers, sample counts, validation frequency, and task-specific selection metric remain *TBD*. These new selection rules are not inferred retrospectively for the historical runs.

B. Networks, Optimization, and Initialization

Table II collects the original manuscript's task-specific settings as candidate starting configurations. The table distinguishes an available configuration from a completed run of Eq. (4). In the new networks, all spiking hidden neurons use the eleven-coefficient definition. The network widths and optimization settings will be frozen before formal evaluation; any subsequent change creates a new configuration and requires a new set of formal runs.

All tasks use cross-entropy and the surrogate in Eq. (26). The supplied S-MNIST and PS-MNIST configuration uses truncated BPTT with chunks of 262 steps. GSC pools hidden spikes by a temporal mean; QTDB optimizes labeled

TABLE II
CANDIDATE BENCHMARK SETTINGS FROM THE ORIGINAL MANUSCRIPT. THESE VALUES PROVIDE THE STARTING CONFIGURATION; THEIR RETENTION AND THE PENDING FIELDS IN SECTION IV-B MUST BE RESOLVED BEFORE THE NEW RUNS. TIMESCALE INTERVALS ARE IN TIMESTEPS.

Item	S-MNIST	PS-MNIST	QTDB	GSC	SHD
Input / output width	1 / 10	1 / 10	4 / 6	120 / 12	700 / 20
Recurrent hidden widths	64, 210	64, 210	31	300	64
Epochs	150	150	250	150	100
Batch size	512	512	128	128	100
Adam learning rate	10^{-2}	10^{-2}	10^{-2}	$5\,10^{-3}$	10^{-2}
StepLR factor / interval	0.1 / 50	0.1 / 50	0.65 / 100	0.5 / 30	0.5 / 20
Fixed threshold θ	1.0	1.0	0.6	0.8	1.0
τ_α	$[2, 60]$	$[2, 60]$	$[20, 120]$	$[10, 80]$	$[10, 80]$
τ_ρ	$[2, 40]$	$[2, 40]$	$[4, 30]$	$[4, 30]$	$[4, 30]$
$\tau_{\eta_0}, \tau_{\eta_1}$	$[0.8, 10]$	$[0.8, 10]$	$[0.8, 8]$	$[0.8, 8]$	$[0.8, 8]$
ω/π	$[0.04, 0.50]$	$[0.04, 0.50]$	$[0.02, 0.20]$	$[0.04, 0.40]$	$[0.04, 0.40]$

timesteps; SHD accumulates readout evidence after warm-up. Exact logit formation for each task, loss weighting or masking, optimizer constants, gradient clipping, weight decay, and the gradient treatment of reset are *TBD*. For truncation, state carryover, graph detachment, and chunk aggregation must be specified separately: carrying a numerical state and carrying its computation graph are different operations.

For each decay $q \in \{\alpha, \rho, \eta_0, \eta_1\}$, a positive time constant τ_q maps to

$$q = \exp(-1/\tau_q), \qquad q_{\text{raw}} = \log \frac{q}{1-q}. \tag{28}$$

Frequency initialization uses the analogous inverse sigmoid of ω/π. The interval alone does not define the sampling distribution; the distribution and neuron-wise sampling rule are *TBD*.

The two rows of $\bar{G}$ are initialized and optimized directly. If a reproducible legacy scalar-reset draw is used, its second row is converted by Eq. (9); the legacy coefficient is discarded after initialization. A direct draw for the new row is also possible. The selected rule, scale, and any regularization remain *TBD*. Controls share that rule and pair common initialization variables where meaningful. Finite-precision treatment of sigmoid endpoints and very small input-vector norms must also be fixed before evaluation.

Formal repetitions use seeds $\{0, 1, 2, 42, 1111\}$. Development runs and their configuration choices are recorded separately. Preliminary runs can enter the formal set only if the predeclared configuration is retained; runs used to alter it become development runs. Infrastructure failures can be retried at the same seed, whereas model numerical failures must be retained in the reported run status. Software versions, hardware, precision, reproducibility settings, and retry criteria for the new runs are *TBD*.

C. Controls with Matched Local Capacity

The structural comparison uses SPRiF, MEMORYRESET5, and REALMODES3 on PS-MNIST and SHD. All three store five state values and learn eleven dynamics coefficients per neuron. They use the same hidden widths, data partitions, readout, training budget, selection rule, and seed-paired input conditions. The comparison tests two specified state organizations, with their scale and conditioning diagnostics.

1) MEMORYRESET5*:* Compute the current spectral state $\mathbf{x}_t^-$, both readouts, the auxiliary state, and the membrane spike as in SPRiF. After firing, additionally update

$$\begin{aligned} \widehat{B} &= \frac{B}{\|B\|_2}, \qquad \gamma_{\text{mem}} = 1, \\ \mathbf{x}_t^+ &= \mathbf{x}_t^- - \gamma_{\text{mem}} \theta z_t \widehat{B}. \end{aligned} \tag{29}$$

The next step uses $\mathbf{x}_t^+$. The current h_t was computed from $\mathbf{x}_t^-$ and receives no extra direct reset. The fixed coefficient adds no learned parameter.

This control introduces one reproducible soft-reset direction through spectral memory. It does not reset all four memory and filter values. Its effect depends on input, state, and readout scales, so the analysis records $\|\Delta \mathbf{x}_t\|_2$, $\|B\|_2$, both readout norms, and the induced drive change. The numerical safeguard for a tiny $\|B\|_2$ is *TBD*. A fixed reset magnitude alone does not establish equal functional perturbation strength across different coordinates.

2) REALMODES3*:* Replace the spectral matrices by

$$A_{\text{real}} = \text{diag}(a_1, a_2, a_3), \qquad B_{\text{real}} = \begin{bmatrix} 1-a_1 \\ 1-a_2 \\ 1-a_3 \end{bmatrix}, \tag{30}$$

where $a_i = \sigma(a_{i,\text{raw}})$ are independently learned. Three real decay parameters replace α, ρ, ω. The auxiliary filter, membrane, scalar reset, and six readout entries are retained.

The first real-mode time constant uses the τ_α interval, and the other two use independent draws from the τ_ρ interval. Distinct poles make the three coordinates controllable, whereas repeated or nearly repeated poles can reduce effective rank or conditioning. Singular values of $[B_{\text{real}}, A_{\text{real}} B_{\text{real}}, A_{\text{real}}^2 B_{\text{real}}]$, with a declared rank tolerance (*TBD*), are reported along with input and readout scales. This control tests real versus real-plus-rotational organization at matched capacity. Setting the original rotation to zero would leave its third coordinate undriven from zero initialization and would test a different, degenerate configuration.

TABLE III
SI-DMS PROTOCOL. AVAILABLE TASK SETTINGS ARE RETAINED FROM THE SUPPLIED SPECIFICATION; UNRESOLVED GENERATOR AND EVALUATION CONSTANTS ARE IDENTIFIED IN THE TEXT.

Item	Setting
Architecture	30 inputs, 64 recurrent hidden neurons, 2 leaky outputs
Training delays	$200, 400, 800, 1600$ ms
Evaluation delays	Training delays and 2500 ms
Models	SPRiF, MEMORYRESET5, ASRNN
Clean training	No forced intervention
Intervention training	$K \in \{0, 1, 2, 4, 8\}$; nominal $f = 15\%$
Primary stress	$K = 40$, nominal $f = 15\%$
Other evaluation counts	$K = 0, 1, 2, 4, 8, 16, 32$
Candidate optimizer	Adam, learning rate 3×10^{-3}
Candidate budget	3000 updates, batch 256, gradient clip 1.0
Rate regularization	Natural-rate target 0.10, weight 0.05
Formal seeds	$0, 1, 2, 42, 1111$ per model and training condition

D. Controlled Baselines and a Full-Method Comparison

TC-LIF [6] and BRF [4] are compared on PS-MNIST and SHD under the common input, partition, readout, validation, and seed protocol. Their native dynamics and initialization are retained and specified in the final configuration. Widths are chosen to bring total trainable parameters as close as practicable to SPRiF, targeting a difference within 5%. Actual widths, parameter counts, native settings, and the achieved budget differences are *TBD*. Equal width is not treated as evidence of equal parameter budget.

DMP-SNN [9] is evaluated on PS-MNIST using its native layer-shared memory organization and full architecture. Compatible data and selection settings are shared; architectural, optimization, and budget differences are stated with the results. This comparison concerns a complete method rather than a neuron substitution. ASRNN [1] is included in SI-DMS with the common recurrent task protocol. The current related-work comparison with PMSN and D-RF concerns equations and state organization; the empirical comparison set is the one specified here.

E. SI-DMS: Task and Spike Intervention

SI-DMS is a generated match/non-match task with one recurrent hidden layer. Thirty input channels form three groups of ten: left cue, right cue, and background noise. Poisson spike trains encode the inputs. A 50-ms pre-period is followed by a 100-ms first cue, a variable delay, and a 100-ms second cue. The two cues define the binary target. Matching cues have the same identity but independent spike realizations. The hidden layer has 64 neurons and connects to a two-class nonspiking leaky readout. Table III gives the candidate training budget and the comparison grid.

At each of K selected delay timesteps, the designated units' membrane candidates are overwritten with a value $v_{\mathrm{set}} > \theta$ immediately before thresholding. Their actual spikes trigger native reset and enter the recurrent pathway. For SPRiF, $\mathbf{x}_t$ and h_t have already been computed normally and are not overwritten. A forced event is distinguished from natural firing when computing rate regularization. The protocol consequently changes membrane voltage, spike generation, reset, and future feedback together.

Clean and intervention training use the same number of updates and generated task conditions. Each formal seed supplies matched evaluation inputs and external intervention schedules across models and across clean/stressed replays. The timestep, cue/noise rates, identity and label sampling law, validation and test sample counts, intervention time sampling law, selected-unit rounding rule, v_{set}, output leak, loss window, and exclusion of forced events from the rate term are *TBD*. In the former task, nominal 15% corresponded to 10 of 64 units; the exact rule for the new runs must be stated rather than inferred from the percentage alone.

For seed s, delay d, count K, and fraction f, the paired loss is

$$\Delta_{s,d}(K, f) = \mathrm{Acc}_{s,d}(0) - \mathrm{Acc}_{s,d}(K, f). \tag{31}$$

Accuracies are percentages, so Δ is measured in percentage points. Each seed is averaged over the five evaluation delays before calculating the across-seed summary. The 2500-ms delay is also reported separately because it lies outside the training set. Counts above eight are outside the intervention-training range. Clean capability, stressed capability, and their paired loss are all required to interpret memory preservation.

F. Initialization, State Diagnostics, and Cost Measurement

Initialization sensitivity uses PS-MNIST with all time-constant intervals multiplied by 0.5, 1, or 2, while holding the frequency interval, readout rule, optimizer, and other settings fixed. The five standard-initialization runs are reused, and the other two ranges each use five formal seeds. The decay transform remains Eq. (28).

The analysis records initialized and learned decays and frequencies, state magnitudes, readout norms, natural firing rates, and numerical failures. A specified tolerance (*TBD*) defines proximity to a parameter boundary. Impulse diagnostics inject a unit current into the fixed-parameter local memory pathway and compare the resulting $\mathbf{x}_t$, h_t, and drive with Eqs. (11)–(15). Checkpoint selection, the displayed-neuron rule, and the observation horizon are *TBD*. Immediate pre/post-reset trajectories test Eq. (24); recurrent clean/stressed replay is treated as a separate network diagnostic.

Cost reporting covers SPRiF, TC-LIF, and BRF on PS-MNIST and SHD. It includes total trainable parameters, dynamics coefficients, persistent state bytes, inference latency, peak device memory, and training time for the stated budget. Timing uses both batch one and the task training batch, with matched device, precision, input length, and software settings. The candidate timing protocol uses 20 warm-up calls and 100 timed repetitions with device synchronization, reporting median and interquartile range. Exact runtime settings and hardware are *TBD*. Full-method DMP-SNN costs are identified with its own configuration. GPU latency and operation counts are not measurements of neuromorphic-device energy.

TABLE IV
HISTORICAL BENCHMARK ACCURACIES (%) FROM THE ORIGINAL MANUSCRIPT. COLUMNS RETAIN THEIR ORIGINAL MODEL IDENTITIES AND REPORTED FIVE-SEED SUMMARIES. A DASH DENOTES AN UNREPORTED ENTRY.

Task	Former Full (13)	Former scalar (12)
S-MNIST	99.28 ± 0.06	—
PS-MNIST	95.86 ± 0.14	95.32 ± 0.18
QTDB	88.43 ± 0.36	87.78 ± 0.30
GSC	94.55 ± 0.31	94.22 ± 0.22
SHD	91.52 ± 0.46	—

TABLE V
SPRIF WITH FIVE STATES AND ELEVEN LEARNED DYNAMICS COEFFICIENTS PER NEURON. ENTRIES REQUIRE NEW MEASUREMENTS; ACCURACY USES FIVE SEEDS.

Task	Accuracy (%)	Total parameters
S-MNIST	*TBD*	*TBD*
PS-MNIST	*TBD*	*TBD*
QTDB	*TBD*	*TBD*
GSC	*TBD*	*TBD*
SHD	*TBD*	*TBD*

G. Metrics and Statistical Units

Sequence accuracy is the primary metric for S-MNIST, PS-MNIST, GSC, SHD, and SI-DMS. QTDB accuracy uses correctly classified labeled timesteps divided by all evaluated labeled timesteps; masking rules are *TBD*. Per-class precision, recall, and macro-F1 are included for QTDB to clarify whether overall accuracy reflects class imbalance; the class mapping and aggregation rule remain *TBD*.

For $S = 5$ independent training seeds with scores a_s, report

$$\bar{a} = \frac{1}{S}\sum_{s=1}^{S} a_s, \qquad s_a = \sqrt{\frac{1}{S-1}\sum_{s=1}^{S}(a_s - \bar{a})^2}. \tag{32}$$

Structural comparisons also report every seed-paired difference and its mean and sample standard deviation. Test examples, delay conditions, and repeated timings are not independent training repetitions. With five seeds, a small difference can remain uncertain; a superiority statement requires evidence beyond a positive point estimate. Published comparator numbers, if included later, will be identified by their original protocol and kept separate from the controlled reruns.

V. RESULTS AND DISCUSSION

A. Evidence Status and Historical Provenance

No new training measurements are available for the eleven-coefficient neuron in Eq. (4). In this manuscript, *TBD* denotes an unavailable result or protocol value. The analytic conclusions in Section III follow from the equations; performance and cost conclusions require the measurements organized below.

The historical Full model stored five values and learned thirteen dynamics coefficients, including a separate auxiliary coupling and a nonmembrane reset coefficient. Its scalar-reset variant retained the independent coupling and learned twelve coefficients. Table IV preserves their reported five-seed benchmark accuracies. The Full-to-scalar change alters the model; the subsequent scaling absorption in Section III-C is a fixed-parameter conversion of the scalar-reset variant. Neither step supplies the outcomes of new training.

Historical SI-DMS also used a different protocol: three seeds and intervention training. At nominal $f = 15\%$, the original Full model reported clean accuracy $100.0 \pm 0.0\%$ and loss 0.2 ± 0.2 percentage points at $K = 40$; the former scalar-reset variant reported $99.4 \pm 0.0\%$ and loss 4.2 ± 0.2 points. These summaries do not populate Table VIII. The original learned-reset, loss-landscape, impulse, and perturbation plots likewise belong to the historical checkpoints. The current manuscript uses an equation-based schematic and reserves empirical diagnostics for the new models.

B. Five-Benchmark Performance

Table V reports the new parameterization at the complete-network level. Total parameters must include all synaptic, bias, dynamics, and output parameters. The five-seed summaries will use the frozen data and checkpoint protocol. Comparison of new and historical accuracy must account for any change in partition, preprocessing, selection, or initialization; such a difference cannot automatically be attributed to removing a reset coefficient.

The empirical interpretation for the five tasks is *TBD*. For QTDB, per-class and macro-F1 results are also *TBD*. These results will determine whether performance is consistent across temporal modalities or concentrated in a particular task regime.

C. Structural Controls and Related Baselines

Table VI organizes the matched-capacity controls and the controlled neuron baselines. The two structural contrasts are SPRiF minus MEMORYRESET5 and SPRiF minus REALMODES3. Every seed-paired accuracy difference is included in Table VII; their means and sample standard deviations are *TBD*. TC-LIF and BRF comparisons require the actual total-parameter matches, while DMP-SNN retains its native architecture.

A difference against MEMORYRESET5 would characterize the tested input-direction spectral reset at its recorded scale. A difference against REALMODES3 would characterize the selected real versus rotational memory organization. These controls address a state-and-coefficient-capacity explanation, while their conditioning and scale measurements address additional confounds. The relevant interpretation remains *TBD*; neither the parameter count nor the kernel derivation establishes task superiority.

D. Spike Intervention and Delay Generalization

Table VIII reports the primary $K = 40$, nominal $f = 15\%$ condition under clean and intervention training. The across-delay summary and the 2500-ms result are shown together. Curves across intervention count and the pre/post-reset state trajectories are *TBD*.

TABLE VI
NEW CONTROLLED COMPARISONS AND THE PS-MNIST FULL-METHOD COMPARISON. THE FIRST THREE MODELS HAVE FIVE STATES AND ELEVEN LOCAL COEFFICIENTS AT THE SAME WIDTH. TC-LIF AND BRF TARGET MATCHED TOTAL PARAMETER BUDGETS; DMP-SNN USES ITS NATIVE ARCHITECTURE. A DASH DENOTES A TASK OUTSIDE THE COMPARISON SCOPE.

Model	PS-MNIST (%)	SHD (%)	PS-MNIST parameters	SHD parameters
SPRiF	*TBD*	*TBD*	*TBD*	*TBD*
MEMORYRESET5	*TBD*	*TBD*	*TBD*	*TBD*
REALMODES3	*TBD*	*TBD*	*TBD*	*TBD*
TC-LIF	*TBD*	*TBD*	*TBD*	*TBD*
BRF	*TBD*	*TBD*	*TBD*	*TBD*
DMP-SNN (native)	*TBD*	—	*TBD*	—

TABLE VII
SEED-PAIRED SPRIF ACCURACY GAINS OVER THE TWO MATCHED-CAPACITY CONTROLS, IN PERCENTAGE POINTS. ALL FORMAL SEEDS ARE SHOWN. THE PAIRED DATA AND NUMERICAL CONDITIONING DIAGNOSTICS ARE REQUIRED FOR INTERPRETATION.

	PS-MNIST		SHD	
Seed	vs. MEMORYRESET5	vs. REALMODES3	vs. MEMORYRESET5	vs. REALMODES3
0	*TBD*	*TBD*	*TBD*	*TBD*
1	*TBD*	*TBD*	*TBD*	*TBD*
2	*TBD*	*TBD*	*TBD*	*TBD*
42	*TBD*	*TBD*	*TBD*	*TBD*
1111	*TBD*	*TBD*	*TBD*	*TBD*

TABLE VIII
SI-DMS AT $K = 40$ AND NOMINAL $f = 15\%$. ACCURACIES ARE PERCENTAGES AND PAIRED LOSSES ARE PERCENTAGE POINTS. EACH DELAY-AVERAGED VALUE IS FORMED WITHIN A SEED, THEN SUMMARIZED OVER FIVE INDEPENDENT TRAINING SEEDS. THE UNSEEN 2500-MS DELAY IS REPORTED SEPARATELY.

		Five-delay mean			2500-ms delay		
Training	Model	Clean	Stress	Δ_{40}	Clean	Stress	Δ_{40}
Clean	SPRiF	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*
Clean	MEMORYRESET5	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*
Clean	ASRNN	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*
Intervention	SPRiF	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*
Intervention	MEMORYRESET5	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*
Intervention	ASRNN	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*

Clean accuracy establishes whether each model can retain the cue before stress. The paired loss measures degradation relative to that capability, and stressed accuracy prevents a low loss at poor clean performance from being interpreted as useful protection. A benefit present under clean training would support a broader interpretation than a benefit restricted to intervention training. Since the intervention changes several connected paths, the result concerns complete model response; attributing all of a difference to an isolated reset path would require an additional path-isolation analysis.

E. Resource Cost and Learned Dynamics

Table IX places the measured neuron comparisons under common execution conditions. Persistent state is counted from the actual layer widths and scalar precision. Latencies are per complete sequence; peak memory and training time are associated with the stated batch and training budget. The table separates storage predicted by the state definition from the measured costs of executing and training the network. DMP-SNN full-method cost and its configuration differences remain *TBD*.

Accuracy and cost must be interpreted together. Additional states require storage even when local updates are linear in width, and dense recurrence can dominate computation. The current theoretical specification supplies neither a latency speedup nor a hardware-energy claim.

The initialization comparison in Table X tests whether a useful operating range extends beyond the original intervals. Boundary fractions, state magnitudes, firing rates, spectral conditioning, and learned drive kernels are *TBD*. The local algebra predicts the kernel for any fixed parameters; observed training behavior must establish which parameter regimes are actually reached.

F. Scope and Limitations

SPRiF restricts intrinsic memory to three spectral coordinates and an auxiliary filter. Learned readouts can cancel modes, and near-boundary parameters can impair conditioning

TABLE IX
RESOURCE MEASUREMENTS FOR THE CONTROLLED NEURON COMPARISONS. STATE STORAGE EXCLUDES PARAMETERS AND BPTT ACTIVATIONS. LATENCY SUMMARIES REQUIRE SYNCHRONIZED TIMINGS UNDER THE SAME DEVICE AND PRECISION. B_{train} IS THE TASK TRAINING BATCH.

Task	Model	Parameters	State (KiB)	Latency (ms)		Peak (MiB)	Train (h)
				Batch 1	B_{train}		
PS-MNIST	SPRiF	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*
PS-MNIST	TC-LIF	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*
PS-MNIST	BRF	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*
SHD	SPRiF	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*
SHD	TC-LIF	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*
SHD	BRF	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*

TABLE X
PS-MNIST INITIALIZATION SENSITIVITY. ALL TIME-CONSTANT INTERVALS ARE SCALED TOGETHER, WITH FREQUENCY AND OTHER SETTINGS FIXED. THE STANDARD-RANGE RESULT REUSES THE MAIN RUNS.

Interval scale	Accuracy (%)	Natural rate	Failed runs
0.5	*TBD*	*TBD*	*TBD*
1	*TBD*	*TBD*	*TBD*
2	*TBD*	*TBD*	*TBD*

within the proved stability range. The resulting family is a constrained temporal realization.

The immediate reset identity does not isolate recurrent feedback or the other changes made by SI-DMS. Interpretation of the controls depends on their reset direction, perturbation scale, conditioning, and initialization. Absorbed scaling preserves fixed-parameter outputs but can change optimization and regularization. Performance, robustness, and resource tradeoffs require the pending measurements.

VI. CONCLUSION

SPRiF couples a real-plus-rotational spectral block and an auxiliary linear filter to a scalar membrane with subtractive reset. The five-state parameterization learns eleven local dynamics coefficients and retains a one-step auxiliary drive. Its equations establish a controllable spectral basis, an explicit effective drive kernel, bounded local states under bounded current, and zero immediate reset increments in all four memory and filter coordinates. The empirical protocol connects these properties to matched-capacity controls, temporal benchmarks, recurrent intervention, and measured resource costs. New measurements are required to determine the practical value of this organization.

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